

Support Vector Machines

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Introduction

- We assume linearly separable data for binary classification.
- Training data $\mathbf{x}_1, \dots, \mathbf{x}_N$, with target variables $t_1, \dots, t_N$, $t_i \in \{-1, 1\}$
- We view the sign of for classification: $h(\mathbf{x}) = \text{sign}(y(\mathbf{x}))$
- Note that we can scale w and b without changing $h(\mathbf{x})$
- Choose w and b so that for the closest point to the decision boundary we satisfy: $y(\mathbf{x}) = \pm 1$
- Therefore we assume:
 - For any nearest point $\mathbf{x}_1$ with $t_1 = -1$, we have $w^T \mathbf{x}_1 + b = -1$
 - For any nearest point $\mathbf{x}_2$ with $t_2 = 1$, we have $w^T \mathbf{x}_2 + b = 1$

Introduction

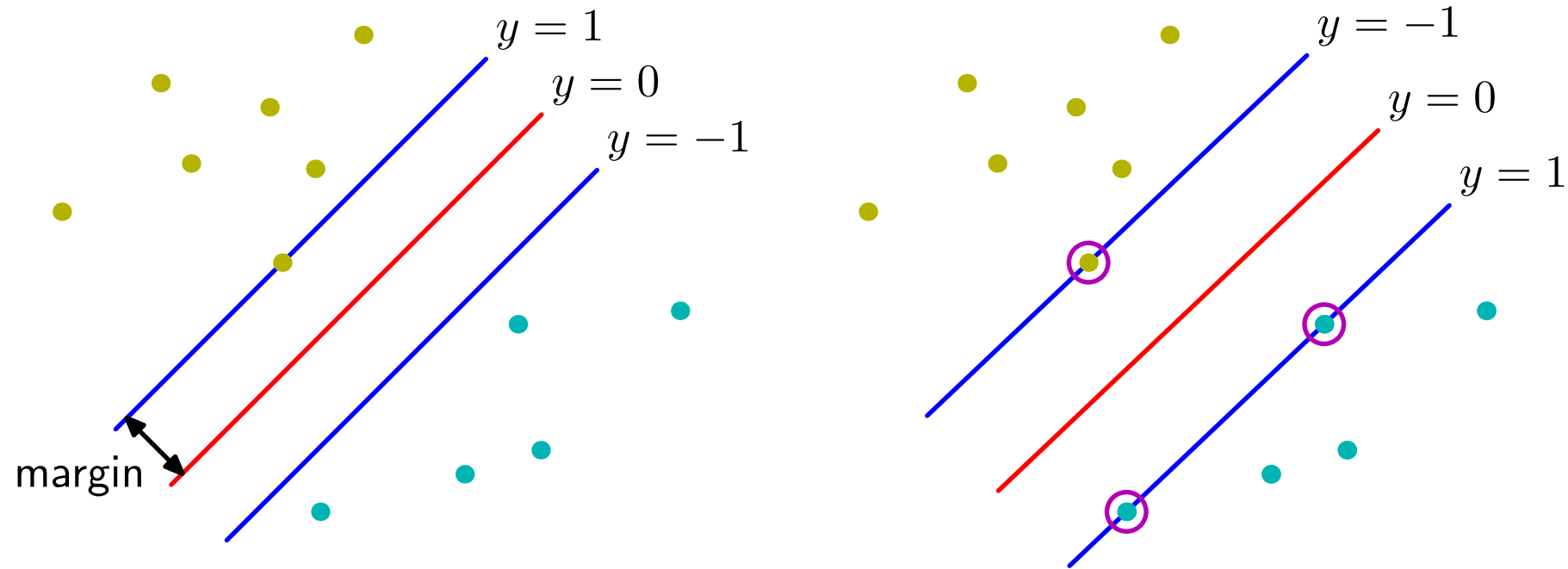
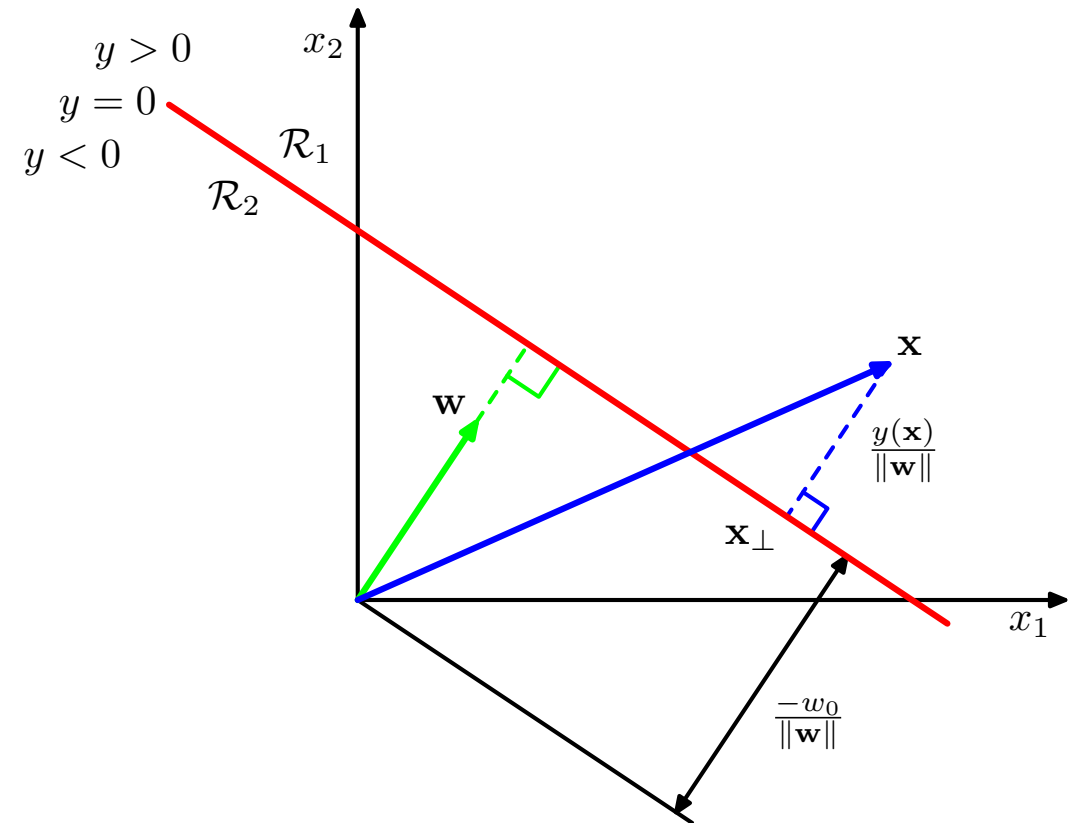


Figure 7.1 from Bishop: The margin is defined as the perpendicular distance between the decision boundary and the closest of the data points, as shown on the left figure. Maximizing the margin leads to a particular choice of decision boundary, as shown on the right. The location of this boundary is determined by a subset of the data points, known as support vectors, which are indicated by the circles.

Review: Geometry of the linear classification

- For a linear predictor function $y(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b$, recall that:
- $\mathbf{w}$ is normal to the decision boundary.
- Distance of $\mathbf{x}$ from decision boundary $y = 0$ is proportional to $y(\mathbf{x})$



SVM Basics

- So, the distance of the data points from $y = 0$ is: $\frac{t_n y_n}{\|\mathbf{w}\|}$
- Thus, for the closet points, this distance (margin) is: $\frac{1}{\|\mathbf{w}\|}$
- To maximise the margin is to minimise $f(\mathbf{w}) = \frac{1}{2} \|\mathbf{w}\|^2$
- Subject to constraints that: $1 \leq t_n y_n$

Primal problem

- For the linear separable SVM, the constrained optimisation is specified

$$\min_{\mathbf{w}, b} \quad f(\mathbf{w}) = \frac{1}{2} \mathbf{w}^\top \mathbf{w} \quad (1)$$

$$\text{subject to} \quad t_i (\mathbf{w}^\top \mathbf{x}_i + b) \geq 1, \quad i = 1, \dots, n \quad (2)$$

- Applying the Karush-Kuhn-Tucker conditions, the primal problem is:

$$L(\mathbf{w}, b, \boldsymbol{\alpha}) = \frac{1}{2} \mathbf{w}^\top \mathbf{w} + \sum_i^n \alpha_i (1 - t_i (\mathbf{w}^\top \mathbf{x}_i + b))$$

$$\alpha_i \geq 0$$

$$\boldsymbol{\alpha} = (\alpha_1 \cdots, \alpha_n)^\top$$

$$1 - t_i (\mathbf{w}^\top \mathbf{x}_i + b) \leq 0$$

$$\alpha_i (1 - t_i (\mathbf{w}^\top \mathbf{x}_i + b)) = 0$$

Slackness conditions

Primal problem

- The primal problem is a convex optimisation problem that can be solved by solvers such as ALGLIB.
- Setting the derivatives of the Lagrangian with respect to the parameters:

$$\frac{\partial L}{\partial b} = 0 \Rightarrow \sum_{i=1}^n \alpha_i t_i = 0$$

$$\frac{\partial L}{\partial \mathbf{w}} = 0 \Rightarrow \mathbf{w} = \sum_{i=1}^n \alpha_i t_i \mathbf{x}_i$$

- We solve an example problem to further illustrate.

From primal to dual problems

- The primal problem is a convex optimization problem that can be solved by solvers such as ALGLIB.
- But our SVM will remain linear ☹️
- Moving towards **dual problem** can be rewarding and can make SVM non-linear 😊 (-- using **kernel** trick!)
- To define the dual problem, we first eliminate the primal variables.

$$\frac{\partial L}{\partial b} = 0 \Rightarrow \sum_{i=1}^n \alpha_i t_i = 0$$

$$\frac{\partial L}{\partial \mathbf{w}} = 0 \Rightarrow \mathbf{w} = \sum_{i=1}^n \alpha_i t_i \mathbf{x}_i$$

Dual problem

- Substitute the results in the primal:

$$\begin{aligned} L(\mathbf{w}, b; \alpha) &= \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j t_i t_j \mathbf{x}_i^\top \mathbf{x}_j + \sum_{i=1}^n \alpha_i \left(1 - t_i \left(\sum_{j=1}^n \alpha_j t_j \mathbf{x}_j^\top \mathbf{x}_i + b \right) \right) \\ &= \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j t_i t_j \mathbf{x}_i^\top \mathbf{x}_j \end{aligned}$$

Dual problem

- We arrive at the following dual problem

$$g(\boldsymbol{\alpha}) = \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} t_i t_j \alpha_i \alpha_j \mathbf{x}_i^\top \mathbf{x}_j$$

$$\alpha_i \geq 0$$

$$\sum_{i=1}^n \alpha_i t_i = 0$$

Complementary slackness condition

- Note that the dual problem is concave (why?) and should be maximized (see the the previous lecture)
- This can be solved by quadratic programming.

Solution for the dual

- Let $\alpha^* = (\alpha_1^*, \dots, \alpha_n^*)^\top$ be the solution for above then

$$\mathbf{w}^* = \sum_{i=1}^n \alpha_i^* t_i \mathbf{x}_i$$

- To determine b^* , we note that for any $\alpha_i^* > 0$

$$t_i(\mathbf{w}^{*\top} \mathbf{x}_i + b^*) = 1$$

Sparsity in predication

- For any test point such as $\mathbf{x}$ the discriminant will be

$$\begin{aligned} y(\mathbf{x}) &= \mathbf{w}^{*\top} \mathbf{x} + b^* \\ &= \sum_{i: \alpha_i^* > 0} \alpha_i^* t_i \mathbf{x}^\top \mathbf{x}_i + b^* \end{aligned}$$

- This shows that prediction uses only a sparse number of $\mathbf{x}_i$'s.
- One interesting point is that only inner products ($\mathbf{x}_i^\top \mathbf{x}_j$) appear in the dual problem and the prediction!
- We can exploit this observation to make SVMs non-linear.